



Computer Network Lab

Lab Task # 8

Name: Muhammad Faisal Farooq

Roll No.:

Class: BS CYS F24

Faculty Name: Ms. Maila Zahra

Quote of the Day

"Technology is a useful servant but a dangerous master." – Christian Lous Lange

--o(Task1)o--

Create VLANs and Assign Ports

➤ **Objective:**

- To divide the switch into two virtual networks for different departments.

➤ **Steps:**

1. Create VLAN 10 (Sales) and VLAN 20 (IT).
2. Assign switch ports fa0/1–2 to VLAN 10 and fa0/3–4 to VLAN 20.
3. Configure fa0/5 as a trunk port to connect the switch to the router.

VLAN:

A VLAN, or virtual local area network, is a way to logically segment a single physical network into multiple smaller virtual networks. This segmentation is done by grouping devices based on function or security needs, rather than their physical location. VLANs improve network performance by reducing broadcast traffic, enhance security by isolating data, and simplify management by allowing flexible grouping without rewiring.

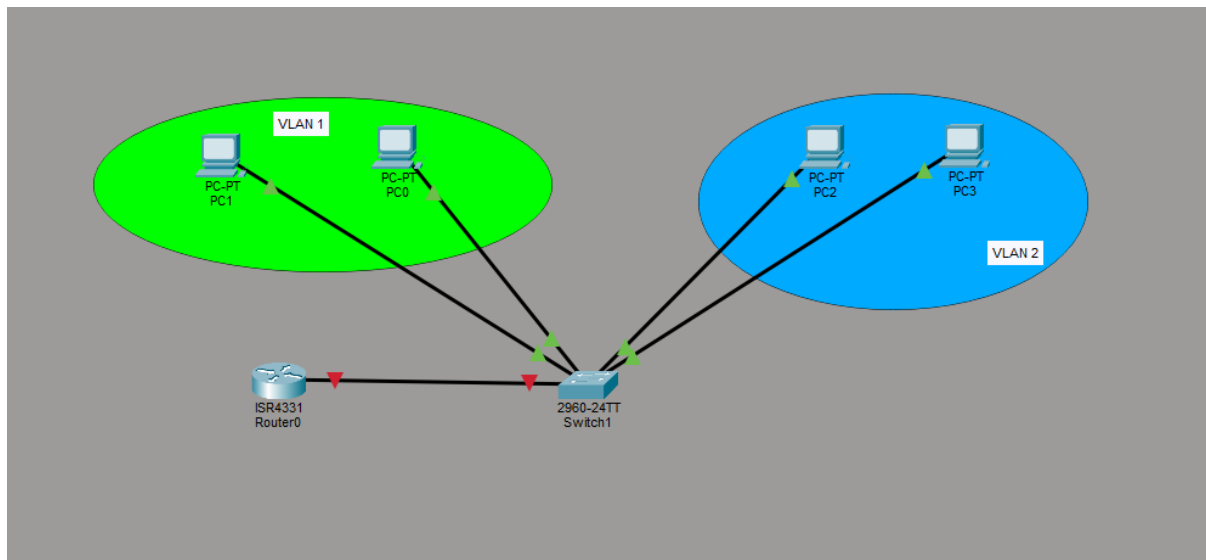


Figure 1: VLAN

How VLANs work:

- **Logical grouping:** Devices are grouped into a VLAN, and devices within the same VLAN can communicate with each other as if they were on a single physical network.
- **Broadcast domains:** Each VLAN is its own broadcast domain. This means that a broadcast frame sent from a device is only received by other devices within that same VLAN, even if they are connected to different switches.
- **VLAN tagging:** Switches use tags to identify which VLAN each frame belongs to, allowing for traffic to be sorted and segregated.
- **Inter-VLAN routing:** Communication between different VLANs requires routing, which can be handled by a router or a Layer 3 switch.

Benefits of using VLANs:

- **Security:** Isolating sensitive data or departments onto separate VLANs improves security by restricting access and limiting who can receive traffic.
- **Performance:** By segmenting the network, VLANs reduce the amount of broadcast traffic that devices must process, which improves overall performance.
- **Flexibility and management:** Administrators can group users by department or function, regardless of their physical location, and easily move or reconfigure devices by simply changing the port's VLAN assignment in the software, without needing to move any cables.

Practical:

- For this simulation we will use 4 PCs to make a VLAN of 2 / 2, a switch and a router.

Note: I will not assigning IPs to PCs just labelling them for ease in future.

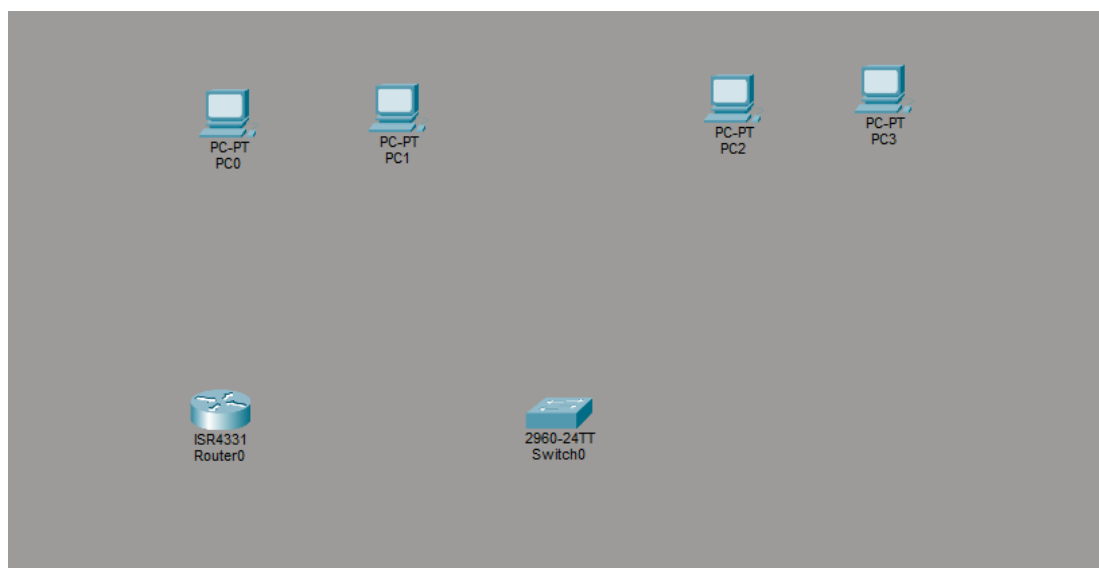


Figure 2: Components in VLAN

- Do not configure the router and PCs we will do it later using terminal.

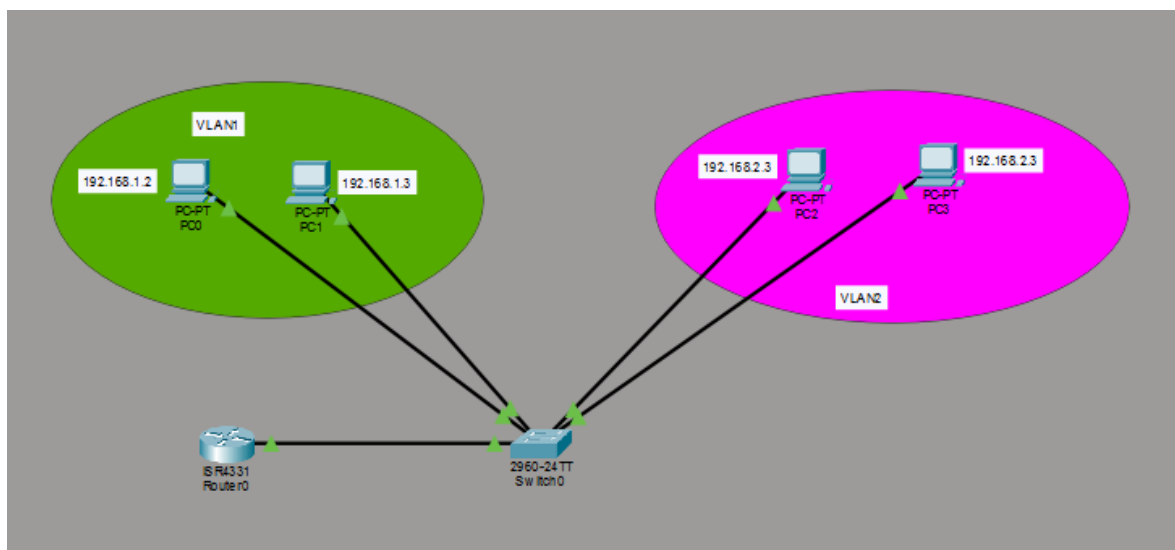


Figure 3: Configured Network

Now creating VLANS:

- Define the VLANs and assign appropriate names (as required).

```
enable
configure terminal

vlan 10
  name Sales
vlan 20
  name IT
exit
```

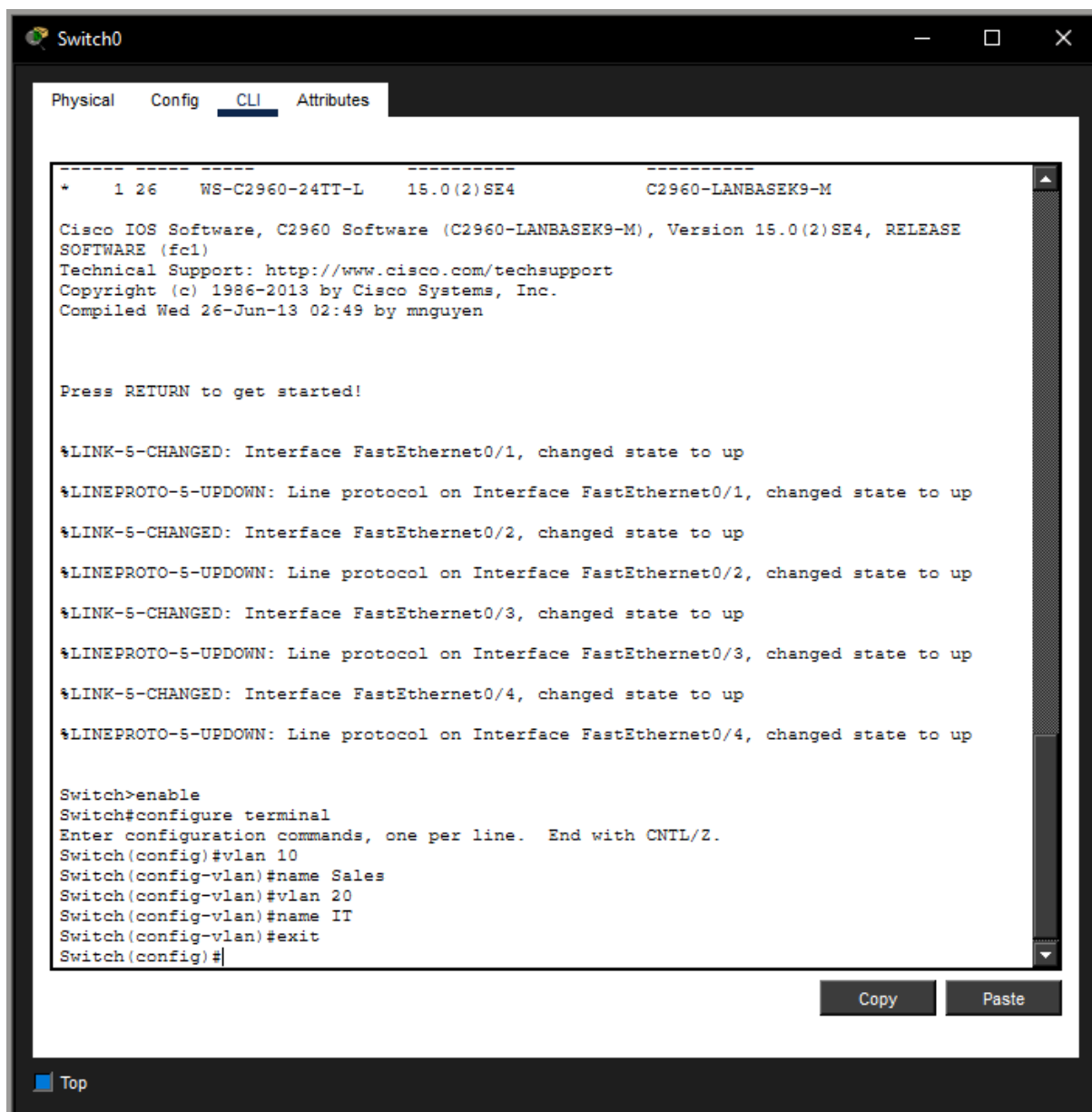


Figure 4: Switch Interface

Assign Access Ports to VLANs:

- Associate the required switch ports with their respective VLANs.

Ports fa0/1–2 = VLAN 10 (Sales):

```
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
exit
```

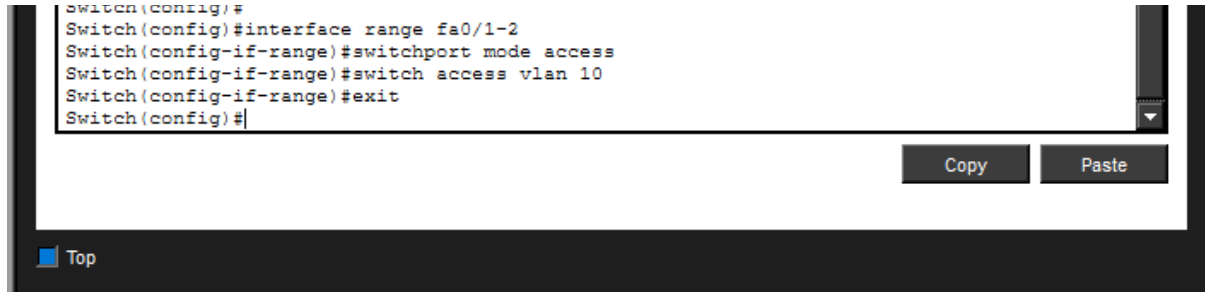


Figure 5: VLAN sales

Ports fa0/3–4 = VLAN 20 (IT):

```
interface range fa0/3 - 4
switchport mode access
switchport access vlan 20
exit
```

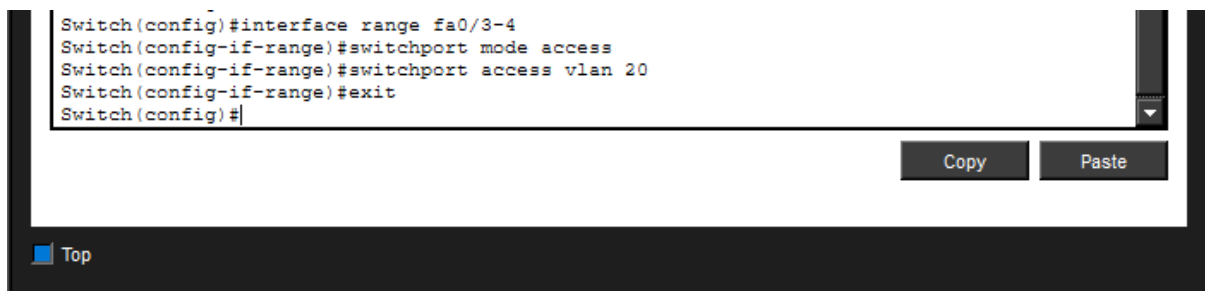


Figure 6: VLAN IT

Configure Trunk Port:

- Set fa0/5 as a trunk port to carry multiple VLANs to the router.

```
interface fa0/5
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

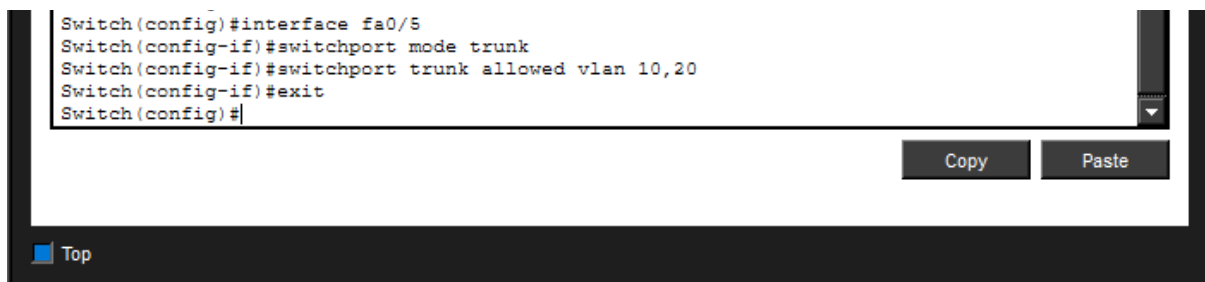


Figure 7: Configure TRUNK mode

Outcome:

- The switch is now divided into two VLANs (Sales and IT), with access ports assigned accordingly. A trunk port is established on fa0/5 to support inter-VLAN routing via the router.

“This separates the network into two broadcast domains.”

Proof:

- By sending PDUs internally and externally of VLANs:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC2	ICMP		0.000	N	0	(e...	
	Successful	PC0	PC1	ICMP		0.000	N	1	(e...	
	Failed	PC3	PC0	ICMP		0.000	N	2	(e...	
	Successful	PC2	PC3	ICMP		0.000	N	3	(e...	

Figure 8: pinging the PCs

--o(The End)o—

--o(Task 2)o--

Configure IP Addresses and Test VLAN Communication

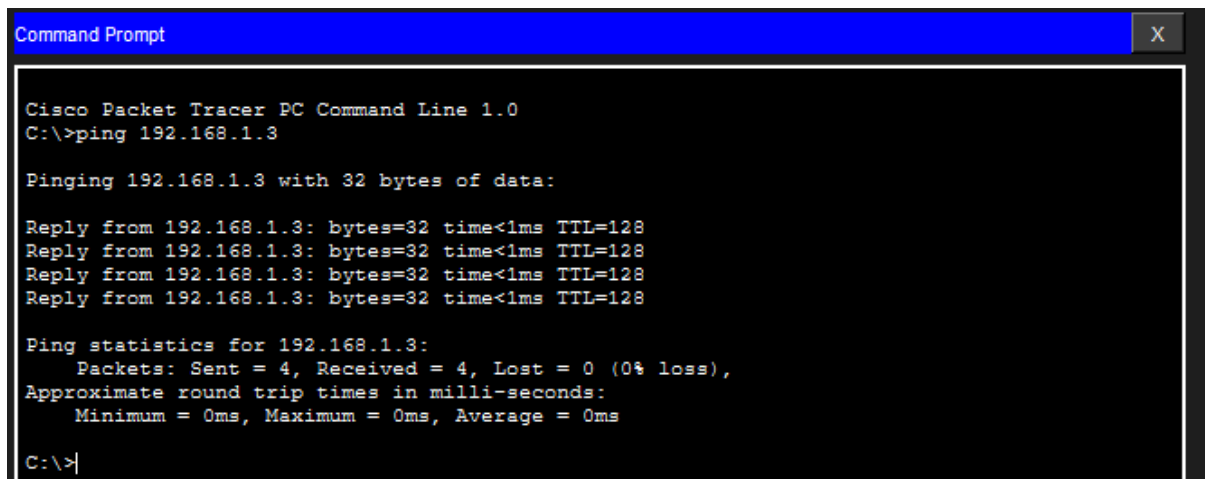
- **Objective:** To assign network identities to PCs and test communication within and between VLANs.

Steps:

- Now assign proper IP addresses to PCs.
- IP configurations:
 - PC0 >> 192.168.1.2
 - PC1 >> 192.168.1.3
 - PC2 >> 192.168.2.2
 - PC3 >> 192.168.2.3

Ping between PCs in the same VLAN:

- Pinging the PCs inside the same VLAN to test the connectivity, It should work.



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

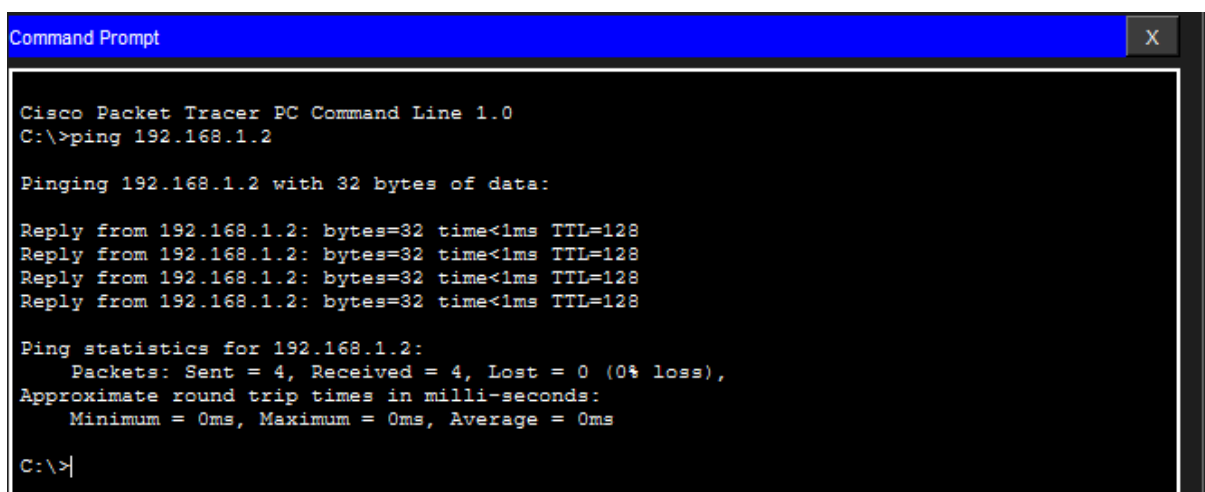
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 9: pinging from PC0 > PC1



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

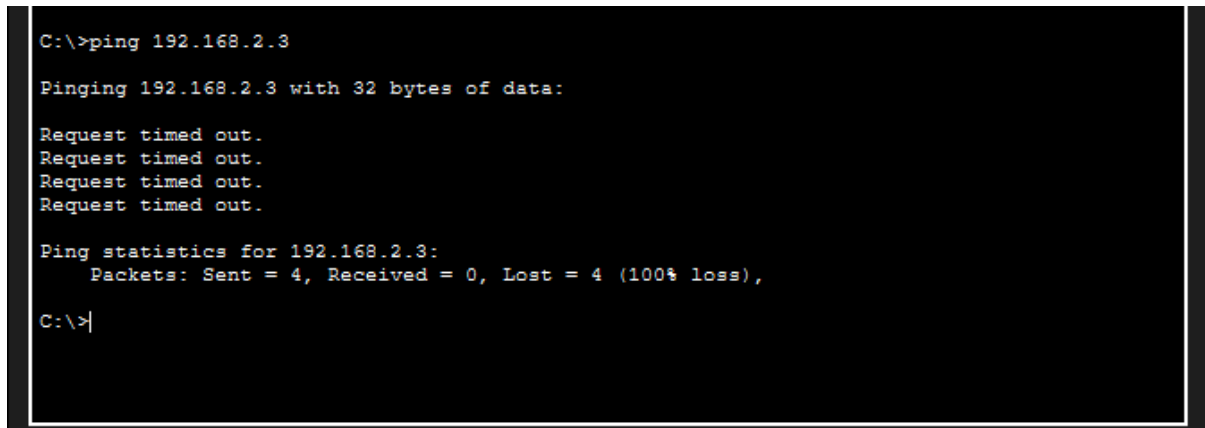
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 10: pinging from PC1 > PC0

Ping between PCs in different VLANs:

- It should fail because routing isn't set up yet.



```
C:\>ping 192.168.2.3

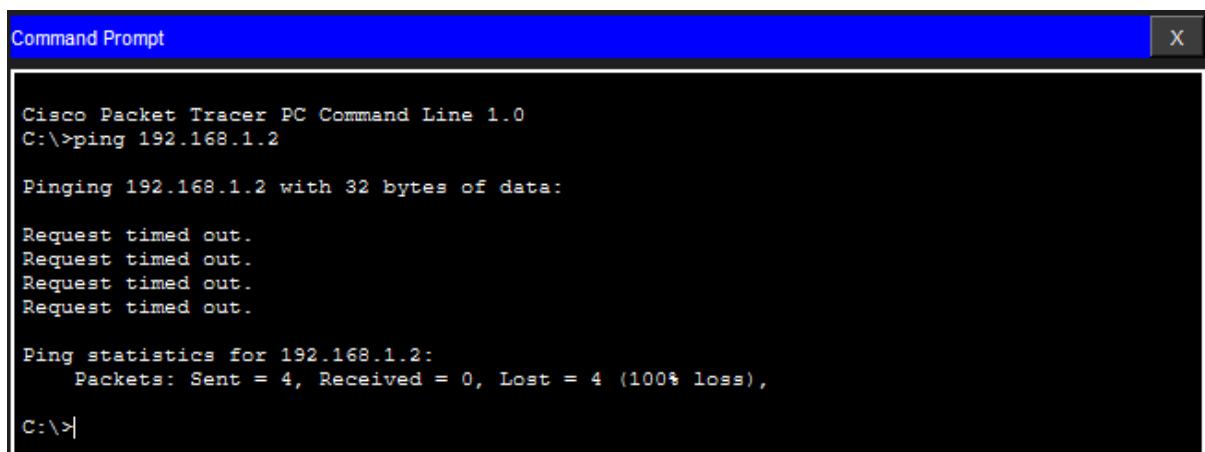
Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Figure 11: pinging from PC0 > PC3



```
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Figure 12: pinging form PC3 > PC0

Inter-VLAN Routing

- The process of moving network traffic between different VLANs using a Layer 3 device (router or Layer 3 switch). Since VLANs create isolated broadcast domains, devices in separate VLANs cannot communicate unless a Layer 3 device routes the traffic between them.

--o(The End)o--

--o(Task 3)o--

Configure Inter-VLAN Routing (Router-on-a-Stick)

Objective:

- To enable communication between VLAN 10 (Sales) and VLAN 20 (IT) using a router.

Steps:

- Configure Router Sub-Interfaces
- Activate the physical interface and create a sub-interface for each VLAN.

```
enable
configure terminal

interface fa0/0
  no shutdown
exit
interface fa0/0.10
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.0
exit

interface fa0/0.20
  encapsulation dot1Q 20
  ip address 192.168.2.1 255.255.255.0
exit
```

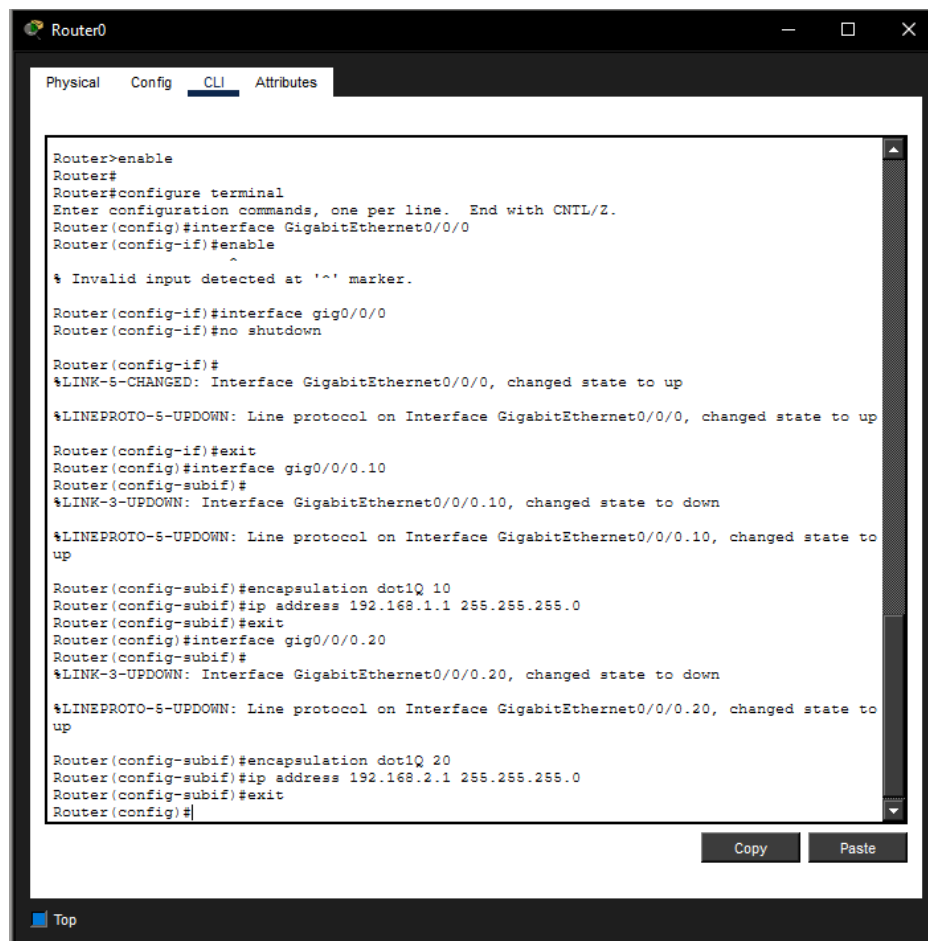


Figure 13: Configuring routing to VLANs

Verify Inter-VLAN Routing:

- After configuration, devices in VLAN 10 and VLAN 20 should be able to ping each other's gateway IPs, and then communicate across VLANs.

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 14: pinging PC1 > PC2 (Inter VLAN)

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 15: Pinging PC3 > PC0 (Inter VLAN)

Outcome:

- The router now performs Inter-VLAN Routing, acting as the Layer 3 gateway for both VLANs. Traffic from VLAN 10 and VLAN 20 can now flow through the router, enabling full communication between the departments.

Hurdles I faced:

- Be sure that you have configured the default gateway for each PC after configuring VLAN.
- It took me some time as I was thinking why its not sending packets in another VLAN, however Upon troubleshooting the entire network host by host I found that I haven't configured the default gateway for each host.

--o(The End)o--